Machine Learning Lab

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Lab 3

Part 1:

1.

```
library(keras); library(keras3); library(tensorflow)
library(ggplot2); library(dplyr); library(tidyr)
library(data.table); library(rpart); library(caret)
library(nnet)
```

```
adults <- readRDS("~/Documents/ML/Labs/Lab3/adults.rds")
adults <- na.omit(adults)
# X_all <- scale(adults[, -which(colnames(adults) == "y")])
X_all <- adults[, -which(colnames(adults) == "y")]
# X_all <- scale(adults[, -6])
y_all <- adults[, "y"]       # labels (0/1)

set.seed(42)
n <- nrow(X_all)
idx <- sample(seq_len(n), size = floor(0.8 * n))
X_train <- X_all[idx, , drop = FALSE]
y_train <- y_all[idx]
X_test  <- X_all[-idx, , drop = FALSE]
y_test  <- y_all[-idx]
```

Based on the information that there are no complex interactions or non-linearities, it can be inferred that a neural network will not excel on this dataset. Neural networks are better suited for capturing non-linear relationships and interactions. Although they can also capture linear relationships, they require more parameters, lack interpretability, and their predictive performance is likely to be close to that of a simple linear regression model.

2.

```r
set.seed(1984)
train_df <- data.frame(y = y_train, X_train)
test_df <- data.frame(y = y_test, X_test)

salary_glm <- glm(y ~ ., data = train_df, family = "binomial")

p_hat <- predict(salary_glm, newdata = test_df, type = "response")


acc_bin <- function(p, y_true, thr = 0.5) mean((p > thr) == y_true)
acc_bin(p_hat, test_df$y, thr = 0.5)
```

```
[1] 0.807964
```

```r
# y_pred <- ifelse(p_hat >= 0.5, 1, 0)
# accuracy <- mean(y_pred == test_df$y)
# accuracy
```

The accuracy is 0.807964.

3.

```r
input_dim <- ncol(X_train)

k_clear_session()
model_A <- keras_model_sequential() |>
  # Hidden layer:
  # units=32 (width of layer), activation="relu" (nonlinearity)
  layer_dense(units = 5, activation = "relu", input_shape = input_dim) |>
  # Output layer for binary classification:
  layer_dense(units = 1, activation = "sigmoid")
```

```r
model_A |>
  compile(
    optimizer = optimizer_adam(1e-3),   # optimizer/learning rate
    loss = "binary_crossentropy",
    metrics = "accuracy"
  )

history_A <- model_A |>
  fit(
    X_train, y_train,
    epochs = 25,
    batch_size = 32,
    validation_split = 0.2,
    verbose = 0
  )

proba_A <- as.numeric(model_A |> predict(X_test, verbose = 0))
acc_A   <- acc_bin(proba_A, y_test)
cat(sprintf("[adults] Model 1 (1 hidden) - test accuracy: %.3f\n", acc_A))
```

```
[adults] Model 1 (1 hidden) - test accuracy: 0.818
```

The accuracy of the MNN is slightly better compared to a standard logistic regression model, but still very close. Moreover, it requires more computational power and parameter tuning, and it lacks interpretability. This is consistent with my assessment in #1.

4.

```r
k_clear_session()
model_D <- keras_model_sequential() |>
  layer_dense(units = 256, activation = "relu", input_shape = input_dim) |>
  layer_dense(units = 128, activation = "relu") |>
  layer_dense(units = 64, activation = "relu") |>
  layer_dense(units = 32, activation = "relu") |>
  layer_dense(units = 1, activation = "sigmoid")

model_D |>
  compile(
    optimizer = optimizer_adam(1e-3),
```

```r
    loss = "binary_crossentropy",
    metrics = "accuracy"
  )

history_D <- model_D |>
  fit(
    X_train, y_train,
    epochs = 25,
    batch_size = 32,
    validation_split = 0.2,
    verbose = 0
  )

proba_D <- as.numeric(model_D |> predict(X_test, verbose = 0))
acc_D   <- acc_bin(proba_D, y_test)
cat(sprintf("[adults] Model 2 (4 hidden) - test accuracy: %.3f\n", acc_D))
```

```
[adults] Model 2 (4 hidden) - test accuracy: 0.828
```

```r
# model_D |> summary()
```

The number of trainable parameters is $1536 + 32896 + 8256 + 2080 + 33 = 44,801$.

The performance is better, but the improvement is not very large. I suspect there may be some limited non-linear relationships and interactions that allow the neural network to capture them, leading to the improvement in accuracy.

5.

```r
adults_aug <- readRDS("~/Documents/ML/Labs/Lab3/adults_aug.rds")
adults_aug <- na.omit(adults_aug)
X_all <- adults_aug[, -which(colnames(adults_aug) == "y")]
y_all <- adults_aug[, "y"]       # labels (0/1)

set.seed(42)
n <- nrow(X_all)
idx <- sample(seq_len(n), size = floor(0.8 * n))
X_train <- X_all[idx, , drop = FALSE]
y_train <- y_all[idx]
```

```r
X_test  <- X_all[-idx, , drop = FALSE]
y_test  <- y_all[-idx]
```

```r
set.seed(1984)
train_df <- data.frame(y = y_train, X_train)
test_df <- data.frame(y = y_test, X_test)

salary_glm <- glm(y ~ ., data = train_df, family = "binomial")

p_hat <- predict(salary_glm, newdata = test_df, type = "response")


acc_bin <- function(p, y_true, thr = 0.5) mean((p > thr) == y_true)
acc_bin(p_hat, test_df$y, thr = 0.5)
```

```
[1] 0.8074521
```

```r
input_dim <- ncol(X_train)

k_clear_session()
model_A <- keras_model_sequential() |>
  # Hidden layer:
  # units=32 (width of layer), activation="relu" (nonlinearity)
  layer_dense(units = 5, activation = "relu", input_shape = input_dim) |>
  # Output layer for binary classification:
  layer_dense(units = 1, activation = "sigmoid")

model_A |>
  compile(
    optimizer = optimizer_adam(1e-3),   # optimizer/learning rate
    loss = "binary_crossentropy",
    metrics = "accuracy"
  )

history_A <- model_A |>
  fit(
    X_train, y_train,
    epochs = 25,
    batch_size = 32,
    validation_split = 0.2,
    verbose = 0
```

```r
  )

proba_A <- as.numeric(model_A |> predict(X_test, verbose = 0))
acc_A   <- acc_bin(proba_A, y_test)
cat(sprintf("[adults_aug] Model 3 (1 hidden) - test accuracy: %.3f\n", acc_A))
```

```
[adults_aug] Model 3 (1 hidden) - test accuracy: 0.965
```

```r
k_clear_session()
model_D <- keras_model_sequential() |>
  layer_dense(units = 256, activation = "relu", input_shape = input_dim) |>
  layer_dense(units = 128, activation = "relu") |>
  layer_dense(units = 64, activation = "relu") |>
  layer_dense(units = 32, activation = "relu") |>
  layer_dense(units = 1, activation = "sigmoid")

model_D |>
  compile(
    optimizer = optimizer_adam(1e-3),
    loss = "binary_crossentropy",
    metrics = "accuracy"
  )

history_D <- model_D |>
  fit(
    X_train, y_train,
    epochs = 25,
    batch_size = 32,
    validation_split = 0.2,
    verbose = 0
  )

proba_D <- as.numeric(model_D |> predict(X_test, verbose = 0))
acc_D   <- acc_bin(proba_D, y_test)
cat(sprintf("[adults_aug] Model 4 (4 hidden) - test accuracy: %.3f\n", acc_D))
```

```
[adults_aug] Model 4 (4 hidden) - test accuracy: 0.996
```

```r
# model_D |> summary()
```

The neural network model should significantly outperform the standard logistic regression model.

On this augmented dataset, the neural network model substantially improves accuracy. This is because many variables containing non-linear relationships and interactions were added to the dataset. Neural networks can capture these effects very well, whereas the standard logistic regression model cannot, and therefore its accuracy is lower than that of the neural network model.

6.

```r
set.seed(1984)

k_clear_session()
model_E <- keras_model_sequential() |>
  layer_dense(units = 256, activation = "relu", input_shape = input_dim) |>
  layer_dropout(rate = 0.1) |>
  layer_dense(units = 128, activation = "relu") |>
  layer_dropout(rate = 0.1) |>
  layer_dense(units = 64, activation = "relu") |>
  layer_dropout(rate = 0.1) |>
  layer_dense(units = 32, activation = "relu") |>
  layer_dense(units = 1, activation = "sigmoid")

model_E |>
  compile(
    optimizer = optimizer_adam(1e-3),
    loss = "binary_crossentropy",
    metrics = "accuracy"
  )

history_E <- model_E |>
  fit(
    X_train, y_train,
    epochs = 25,
    batch_size = 32,
    validation_split = 0.2,
    verbose = 0
  )

proba_E <- as.numeric(model_E |> predict(X_test, verbose = 0))
```

```r
acc_E   <- acc_bin(proba_E, y_test)
cat(sprintf("[adults_aug] Model 5 (4 hidden) - test accuracy: %.3f\n", acc_E))
```

```
[adults_aug] Model 5 (4 hidden) - test accuracy: 0.997
```

```r
# model_E |> summary()

# results <- model_E |> evaluate(X_test, y_test, verbose = 0)
# results$accuracy
```

```r
k_clear_session()
model_F <- keras_model_sequential() |>
  layer_dense(units = 256, activation = "relu", input_shape = input_dim) |>
  layer_dropout(rate = 0.9) |>
  layer_dense(units = 128, activation = "relu") |>
  layer_dropout(rate = 0.9) |>
  layer_dense(units = 64, activation = "relu") |>
  layer_dropout(rate = 0.9) |>
  layer_dense(units = 32, activation = "relu") |>
  layer_dense(units = 1, activation = "sigmoid")

model_F |>
  compile(
    optimizer = optimizer_adam(1e-3),
    loss = "binary_crossentropy",
    metrics = "accuracy"
  )

history_F <- model_F |>
  fit(
    X_train, y_train,
    epochs = 25,
    batch_size = 32,
    validation_split = 0.2,
    verbose = 0
  )

proba_F <- as.numeric(model_F |> predict(X_test, verbose = 0))
acc_F   <- acc_bin(proba_F, y_test)
cat(sprintf("[adults_aug] Model 6 (4 hidden) - test accuracy: %.3f\n", acc_F))
```

```
[adults_aug] Model 6 (4 hidden) - test accuracy: 0.534
```

```
# model_F |> summary()
```

A dropout rate of 0.1 performs better, possibly because dropout learning provides regularization and reduces variance. Although it may increase bias, the reduction in variance outweighs the increase in bias, which makes it worthwhile, and accuracy is further improved.

With a dropout rate of 0.9, accuracy drops dramatically. This is likely also due to reduced variance but with a much greater increase in bias. The increase in bias clearly outweighs the benefits of reduced variance, leading to a sharp decline in accuracy.

Part 2

1.

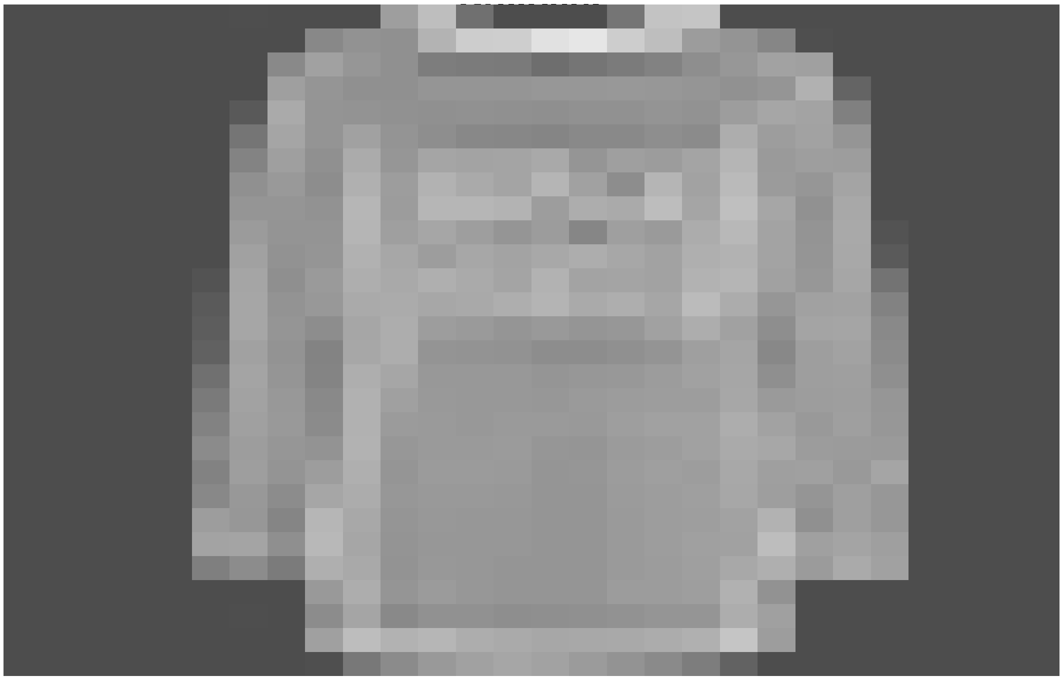
```
fashion_train <- readRDS("~/Documents/ML/Labs/Lab3/fashion_2016_train.rds")
fashion_test <- readRDS("~/Documents/ML/Labs/Lab3/fashion_2016_test.rds")
```

2.

```
x_train <- fashion_train$images
y_train <- fashion_train$labels
x_test <- fashion_test$images
y_test <- fashion_test$labels


# fashion_train$labels
num <- 5
# fashion_label <- fashion_train$labels[num]
# fashion_train$fashion_labels[fashion_label+1]

par(mar = c(0,0,0,0))
test <- x_train[num,,,]
item_name <- fashion_train$fashion_labels[fashion_train$labels[num]+1]
image(t(apply(test, 2, rev)), col = gray.colors(256), axes = FALSE,
      main = item_name)
```



```r
print(item_name)
```

```
[1] "Pullover"
```

```r
img_rows <- 28L; img_cols <- 28L; channels <- 1L
x_train <- array_reshape(x_train, c(nrow(x_train), img_rows, img_cols, channels)) # Note: onl
x_test  <- array_reshape(x_test,  c(nrow(x_test),  img_rows, img_cols, channels))
# x_train <- x_train / 255 # normalize by dividing by max-value (1=most dark; 0=most bright)
# x_test  <- x_test  / 255
y_train_cat <- to_categorical(y_train, num_classes = 10)  # one-hot targets
y_test_cat  <- to_categorical(y_test,  num_classes = 10)
```

```r
k_clear_session()
cnn <- keras_model_sequential() |>
  layer_conv_2d(filters = 8, kernel_size = c(3,3), activation = "relu",
                input_shape = c(img_rows, img_cols, channels)) |>
  layer_max_pooling_2d(pool_size = c(2,2)) |>
  layer_flatten() |>
  # layer_dropout(rate = 0.95) |>
  layer_dense(units = 8, activation = "relu") |>
  layer_dense(units = 10, activation = "softmax")
```

```r
set.seed(1984)
cnn |>
  compile(
    optimizer = optimizer_adam(),
    loss      = "sparse_categorical_crossentropy",
    metrics   = "accuracy"
  )

history_cnn <- cnn |>
  fit(
    x_train, y_train,
    epochs = 3,
    batch_size = 128,
    validation_split = 0.1,
    verbose = 0
  )

# Evaluate on test set
cnn_eval <- cnn |>
  evaluate(x_test, y_test, verbose = 0)
cat(sprintf("\n[fashion] CNN - test accuracy: %.3f\n", as.numeric(cnn_eval['accuracy'])))
```

```
[fashion] CNN - test accuracy: 0.826
```

The accuracy is 0.826.

Increasing M (number of hidden layers), K (number of convolutional layers), the number of filters, or the number of hidden units in the fully connected layer should improve performance, as it allows the model to capture more local features.

3.

```r
k_clear_session()
cnn_complex <- keras_model_sequential() |>
  layer_conv_2d(filters = 32, kernel_size = c(3,3), activation = "relu",
                input_shape = c(img_rows, img_cols, channels)) |>
  layer_max_pooling_2d(pool_size = c(2,2)) |>
  layer_conv_2d(filters = 64, kernel_size = c(3,3), activation = "relu",
                input_shape = c(img_rows, img_cols, channels)) |>
```

```r
  layer_max_pooling_2d(pool_size = c(2,2)) |>
  layer_flatten() |>
  # layer_dropout(rate = 0.95) |>
  layer_dense(units = 64, activation = "relu") |>
  layer_dense(units = 32, activation = "relu") |>
  layer_dense(units = 10, activation = "softmax")

cnn_complex |>
  compile(
    optimizer = optimizer_adam(),
    loss      = "sparse_categorical_crossentropy",
    metrics   = "accuracy"
  )

history_cnn_complex <- cnn_complex |>
  fit(
    x_train, y_train,
    epochs = 3,
    batch_size = 128,
    validation_split = 0.1,
    verbose = 0
  )

# Evaluate on test set
cnn_complex_eval <- cnn_complex |>
  evaluate(x_test, y_test, verbose = 0)
cat(sprintf("\n[fashion] cnn_complex - test accuracy: %.3f\n", as.numeric(cnn_complex_eval['a
```

```
[fashion] cnn_complex - test accuracy: 0.862
```

The accuracy decreased, likely due to overfitting, as no regularization was applied.

From the perspective of the bias-variance trade-off, the model achieved lower bias, but the variance increased too much, which led to the drop in performance.

```r
cnn |> summary()
```

```
Model: "sequential"

 Layer (type)                      Output Shape                  Param #
```

```
 conv2d (Conv2D)                    (None, 26, 26, 8)                   80

 max_pooling2d (MaxPooling2D)       (None, 13, 13, 8)                    0

 flatten (Flatten)                  (None, 1352)                         0

 dense (Dense)                      (None, 8)                       10,824

 dense_1 (Dense)                    (None, 10)                          90

 Total params: 32,984 (128.85 KB)
 Trainable params: 10,994 (42.95 KB)
 Non-trainable params: 0 (0.00 B)
 Optimizer params: 21,990 (85.90 KB)
```

```
cnn_complex |> summary()
```

```
Model: "sequential"

 Layer (type)                       Output Shape                   Param #

 conv2d (Conv2D)                    (None, 26, 26, 32)                 320

 max_pooling2d (MaxPooling2D)       (None, 13, 13, 32)                   0

 conv2d_1 (Conv2D)                  (None, 11, 11, 64)              18,496

 max_pooling2d_1 (MaxPooling2D)     (None, 5, 5, 64)                     0

 flatten (Flatten)                  (None, 1600)                         0

 dense (Dense)                      (None, 64)                     102,464

 dense_1 (Dense)                    (None, 32)                       2,080

 dense_2 (Dense)                    (None, 10)                         330

 Total params: 371,072 (1.42 MB)
 Trainable params: 123,690 (483.16 KB)
 Non-trainable params: 0 (0.00 B)
 Optimizer params: 247,382 (966.34 KB)
```

4.

Because downsampling increases bias and reduces variance. However, if the downsampling is too aggressive, too much detail will be lost, making it difficult for the filters to effectively capture features.

5.

```
predictions <- cnn %>% predict(x_test)
```

```
289/289 - 0s - 739us/step
```

```
predicted_classes <- apply(predictions, 1, which.max)- 1
accuracy_by_class_dt <- data.table(actual=y_test,predicted=predicted_classes)
accuracy_by_class_dt[,correct := fifelse(actual==predicted, yes = 1, no = 0)]
fashion_labels <- c("T-shirt/top", "Trouser", "Pullover", "Dress", "Coat",
"Sandal", "Shirt", "Sneaker", "Bag", "Ankle boot")
fashion_labels_dt <- data.table(label=fashion_labels,number=0:9)
accuracy_by_class_dt <- merge(x=accuracy_by_class_dt,y=fashion_labels_dt,by.x='actual',by.y=
accuracy_by_class_dt[,.(accuracy=mean(correct)),by=label][order(accuracy,decreasing = T)]
```

```
          label  accuracy
         <char>     <num>
 1:     Trouser 0.9523810
 2:  Ankle boot 0.9368879
 3:         Bag 0.9277487
 4:     Sneaker 0.9084132
 5: T-shirt/top 0.8812785
 6:      Sandal 0.8806134
 7:       Dress 0.8743287
 8:    Pullover 0.7379619
 9:        Coat 0.7252066
10:       Shirt 0.4453871
```

From the table above, we can see that there are significant differences in predictability across categories. For example, Ankle boot, Sandal, Bag, and Trouser are the easiest to classify correctly, whereas Shirt is difficult to classify.

This can be considered reasonable.

6.

```r
fashion_unlabeled <- readRDS("~/Documents/ML/Labs/Lab3/fashion_2015_2016_unlabeled.rds")

x_test <- fashion_unlabeled$images

predictions <- cnn %>% predict(x_test)
```

```
2188/2188 - 1s - 581us/step
```

```r
predicted_classes <- apply(predictions, 1, which.max)- 1
fashion_unlabeled$labels <- predicted_classes

demo_dt <- as.data.table(fashion_unlabeled$demographics)
demo_dt[, id := .I]

labels_dt <- data.table(id = seq_along(fashion_unlabeled$labels),
                        labels = fashion_unlabeled$labels)

merged_dt <- merge(demo_dt, labels_dt, by = "id")

# fashion_labels_dt

fashion_full_dt <- merge(
  x    = merged_dt,
  y    = fashion_labels_dt,
  by.x = "labels",
  by.y = "number",
  all.x = TRUE
)
```

```r
# fashion_full_dt
```

```r
fashion_full_dt |>
  group_by(label, year) |>
  summarize(
    age_average = mean(age),
    income_average = mean(income),
    education_years_average = mean(education_years)
  )
```

```
`summarise()` has grouped output by 'label'. You can override using the
`.groups` argument.
```

```
# A tibble: 20 x 5
# Groups:   label [10]
   label        year age_average income_average education_years_average
   <chr>       <dbl>       <dbl>          <dbl>                   <dbl>
 1 Ankle boot   2016        35.2         71023.                    15.6
 2 Ankle boot   2017        35.2         71876.                    15.6
 3 Bag          2016        35.5         77249.                    15.3
 4 Bag          2017        35.3         77200.                    15.3
 5 Coat         2016        39.2         64347.                    14.2
 6 Coat         2017        39.3         64515.                    14.2
 7 Dress        2016        39.2         54622.                    13.9
 8 Dress        2017        39.7         54482.                    13.9
 9 Pullover     2016        36.1         54982.                    13.9
10 Pullover     2017        36.0         55873.                    13.9
11 Sandal       2016        35.1         60298.                    14.5
12 Sandal       2017        35.3         61101.                    14.5
13 Shirt        2016        36.1         61509.                    14.5
14 Shirt        2017        35.8         61537.                    14.4
15 Sneaker      2016        31.5         65749.                    15.1
16 Sneaker      2017        31.8         66287.                    15.1
17 T-shirt/top  2016        35.6         50189.                    13.4
18 T-shirt/top  2017        35.7         49765.                    13.5
19 Trouser      2016        35.4         48707.                    13.4
20 Trouser      2017        35.4         48642.                    13.4
```

Consumer demographics remained largely stable between 2016 and 2017 across all product categories. Average age, income, and education levels changed only slightly, with differences too small to be meaningful.

```r
fashion_full_dt$label <- factor(fashion_full_dt$label)
levels(fashion_full_dt$label)
```

```
 [1] "Ankle boot"  "Bag"         "Coat"        "Dress"       "Pullover"
 [6] "Sandal"      "Shirt"       "Sneaker"     "T-shirt/top" "Trouser"
```

```r
fashion_full_dt$label <- relevel(fashion_full_dt$label, ref = "Bag")

mn_fit <- multinom(label ~ age*year + ses_score*year, data = fashion_full_dt, trace = FALSE)
```

```r
# summary(mn_fit)

exp(coef(mn_fit))
```

```
            (Intercept) age      year ses_score  age:year year:ses_score
Ankle boot    1.0000001   1 1.0002092 0.9999999 0.9999990      0.9997831
Coat          0.9999998   1 0.9996478 0.9999998 1.0000145      0.9996144
Dress         0.9999998   1 0.9996916 0.9999997 1.0000153      0.9993296
Pullover      1.0000001   1 1.0001528 0.9999997 1.0000030      0.9993160
Sandal        1.0000001   1 1.0002122 0.9999997 0.9999994      0.9994675
Shirt         1.0000000   1 0.9999801 0.9999998 1.0000027      0.9995035
Sneaker       1.0000004   1 1.0007723 0.9999998 0.9999841      0.9996306
T-shirt/top   1.0000001   1 1.0001823 0.9999996 1.0000012      0.9991334
Trouser       1.0000000   1 1.0000864 0.9999995 1.0000006      0.9990874
```

In the multinomial logistic regression, I set bags as the reference category. Across all other product categories, the odds ratios for age, socio-economic status (SES), and year were very close to 1, suggesting that these factors had no meaningful effect on the likelihood of choosing a given product over bags. In addition, the interaction terms ($age \times year$ and $SES \times year$) were also close to 1, indicating that the influence of age and SES on product choice did not change between 2016 and 2017. Overall, consumer behavior patterns remained stable across the two years.